

Cracks in Steel Bridges (CSB) dataset: data underlying the publication: Deep Learning for Segmentation of Cracks in High-Resolution Images of Steel Bridges

DOI: 10.4121/6162a9b6-2a20-4600-8207-e9dcd53a264a.v1

Cite

DATASET

Version 1 (old) ▼

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The presented dataset used for the experiments is described in the article "Deep Learning for Segmentation of Cracks in High-Resolution Images of Steel Bridges" (doi:https://doi.org/10.48550/arXiv.2403.17725). The dataset consists of images of steel bridge structures and pixel-wise fatigue crack annotations. Some of the images contain bridge structures with cracks or corrosion, while others capture structures without any defect. The images are provided by bridge infrastructure owners "Rijkswaterstaat" and "ProRail" and by "Nebest" engineering company. The annotation of images was made using a semi-automatic annotation tool described in the article "Segmentation Tool for Images of Cracks" (doi:https://doi.org/10.1007/978-3-031-35399-4_8) and which implementation is available at https://github.com/akomp22/crack-segmentation-tool. The dataset consists of high-resolution images and is stored in the folder "entire images". The images are divided into test and train sets. Images that capture cracks are stored in the folder "crack_train" and "crack_test". Images capturing structure without a crack are stored in folders "nocrack_train" and "nocrack_test". For each image, a .json file is stored in the same folder and under the same name as the corresponding image. The .json file stores the position (x,y) of pixels on the image, which lie in a crack region. An example of a code to generate a binary segmentation map from the .json files is given in the "read_json_annotation.py" file. Additional patch datasets were generated from the entire images. The patch datasets are stored in the "patch dataset" folder. The multiple patch datasets differ by the patch size, number of patches, and fraction of patches that do not contain cracks among all patches of the particular dataset. For more explanations, please refer to the article: https://doi.org/10.48550/arXiv.2403.17725



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History

2024-04-16 first online, published, posted

Publisher

4TU.ResearchData

Format

images/jpeg annotations/json application/python

References

https://doi.org/10.1007/978-3-031-35399-4_8
 https://doi.org/10.48550/arXiv.2403.17725

Organizations

TU Eindhoven, Department of the Built Environment

DATA

Files (3)

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